

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 6

Status: active BA1 execution plan after BA1-T1 candidate derivation.

Supersedes: Revision 5 only for forward execution state; R1-R5 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED by BA0-T3.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.
- No historical focus set, systems-modeling metaclass or threat-method element is inherited automatically into BA1.

2. BA0 boundary carried forward

Base Analysis must preserve baseline-scoped shared semantic identity, source/origin provenance and explicit unresolved state sufficiently for reusable human/method projections, source drill-down, change-impact reasoning and source-localized feedback handoff.

The representation remains methodology-neutral and implementation-independent. Grounded, derived and diagnostic/unresolved are responsibility-level origin/state distinctions; a generic reviewed analytical-addition class is not required.

3. BA1 objective

BA1 decides the **minimal first-class identity ontology** needed to realize the BA0 boundary.

For every recurring semantic responsibility, test first whether it can remain:

- relation;
- role/classification;
- property/value;
- constraint/state;
- provenance/diagnostic record;
- derived/projection construct.

Only independent semantic identity plus reusable subtype-specific invariants justify a first-class split.

```
semantic responsibility exists
    !=
first-class metaclass required
```

4. BA1-T1 - minimal ontology candidate derivation

Status: COMPLETED / PROVISIONAL CANDIDATE.

BA1-T1 rejects both extremes:

1. one undifferentiated analytical element type is too weak because it hides the distinction between identity of a project referent and identity/origin/evolution of analytical claims;
2. domain-specific root taxonomies are premature because current evidence does not yet force separate Behavior, Information, Participant, Store, Contract, Boundary, State, etc. metaclasses.

The resulting falsifiable candidate is:

BAReferent CANDIDATE / NOT ACCEPTED
BAProposition CANDIDATE / NOT ACCEPTED

BAReferent provides independent identity for shared project meaning. BAProposition provides independent identity for provenance-bearing shared analytical claims. Specific semantic kinds remain open split hypotheses.

No relation vocabulary is closed by T1.

5. BA1-T2 - split-or-collapse pressure test

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

Pressure-test the two-family candidate against deliberately difficult cases:

1. **Behavior/event identity:** can FR-3.4 delivery and the physical handoff milestone be represented without a dedicated Behavior/Event type while preserving specialized constraints and cross-branch reuse?
2. **Information/resource lifecycle:** do RecognitionCapture, OrderEvaluation, Reservation, result objects and their correlation/lifecycle semantics require a separate Information/Resource type?
3. **Responsibility and externality:** can internal/external ownership, provider boundaries and capability/service consumption remain propositions/roles over referents without dedicated Participant/Component/Boundary roots?
4. **Store and contract identity:** do the store candidate InventoryLedger and the contract candidates PaymentProviderContract / CarrierContract require subtype-specific first-class invariants?
5. **State/context:** can ACCEPTED, authorized/declined/indeterminate, reservation lifecycle and failure states remain values/qualifiers/propositions?
6. **Method consumer:** can one bounded method-specific projection select what it needs without reconstructing missing domain kinds independently from raw prose?
7. **Collapse test:** can BAREferent and BAProposition be merged without losing assertion-level provenance/change semantics?

BA1-T2 exit

Record for every deferred split one of:

- NOT FORCED;
- FORCED CANDIDATE SPLIT;
- REQUIRES MORE EVIDENCE;
- REJECTED.

Do not close BA1 merely because the two-family candidate survives one corpus fragment.

6. BA1 closure review

After BA1-T2, execute a distinct closure review only if the candidate is stable enough to define a minimal ontology without embedding relation vocabulary or provenance mechanics.

Closure must state:

- accepted first-class BAE types/families;
- explicit non-types/roles/properties/deferred constructs;
- first-class identity criterion;
- reopen criteria;
- what remains for BA2-BA6.

Only explicit BA1 closure authorizes BA2.

7. Sequence after BA1 closure

Do not skip phases:

BA2 - Relations and canonical action vocabulary
BA3 - Document-to-BAE derivation, provenance and authority
BA4 - Multi-level human and method projections
BA5 - Controlled vocabulary and optional authoring/extraction assistance
BA6 - Base Analysis closure and regression

BA2

Define the smallest stable relation/action vocabulary over the accepted BA1 identity ontology. Keep semantic relation, canonical lexical label and authoring synonym assistance separate.

BA3

Close identity/equivalence/lifecycle rules, source locators, grounded/derived/diagnostic materialization, acceptance/rejection, stale state and document-to-analysis traceability.

BA4

Define derived human and method projections. No view is source authority.

BA5

Define controlled vocabulary and optional assistance. LLM/NLP proposals remain candidate-producing support and cannot establish authority silently.

BA6

Regress the Base Analysis against closed corpora and at least one structurally different holdout before formal overlay work.

8. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The BA0-T2 STRIDE probe remains only a pre-overlay consumer probe.

9. Next authorized microstep

Execute only:

BA1-T2 - split-or-collapse pressure test of the two-family candidate.

Do not start BA2, formal STRIDE overlay design, Common Finding schema or implementation work until BA1 closes explicitly.